

Final examination of Introduction to Networking

Christian Rinderknecht

19 June 2007

1. **Questions.** Suppose two hosts A and B , separated by 1,000 km, and connected by a direct link of $R = 1$ Mbps. Propagation speed of the link is 2.5×10^8 m/sec.
 - (a) Calculate the *bandwidth-delay product* $R \times d_{\text{prop}}$.
 - (b) Consider sending a file of 400,000 bits from host A to host B . Suppose the file is sent continuously as one big message. What is the maximum number of bits that we will be in the link at any given time?
 - (c) Provide an interpretation of the bandwidth-delay product.
 - (d) What is the width (in meters) of a bit in the link?
2. **Question.** Referring to question 1, suppose we can modify R . For what value of R is the width of a bit as long as the length of the link?
3. **Question.** What are the advantages of message segmentation in packet-switched networks? What are the disadvantages?
4. **Question.** Is HFC bandwidth dedicated or shared among users? Are collisions possible in a downstream HFC channel?
5. **Question.** List five tasks that a protocol layer can perform. Is it possible that one (or more) of these tasks could be performed by two (or more) layers?
6. **Questions.** True or false?
 - (a) Suppose a user requests a Web page that consists of some text and two images. For this page the client will send one request and receive three response messages.
 - (b) Two distinct Web pages can be sent over the same persistent connection.
 - (c) With non-persistent connections between browser and origin server, it is possible for a single TCP segment to carry two distinct HTTP request messages.
 - (d) The **Date:** header in the HTTP response message indicates when the object in the response was last modified.